



Privacy fatigue and its effects on ChatGPT acceptance among undergraduate students: is privacy dead?

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Received: 9 September 2024 / Accepted: 25 November 2024 / Published online: 10 January 2025
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Abstract

OpenAI’s generative pre-trained transformer (ChatGPT) is rapidly transforming fields such as education and becoming an integral part of our daily lives. However, its rise has sparked intense privacy debates, raising concerns about the storage of personal information and attracting global regulatory scrutiny. ChatGPT is gaining popularity among undergraduate students owing to its personalized learning capabilities, leading many universities to provide guidelines and seminars on its use. Curiosity about the reasons for ChatGPT’s popularity inspired our research. Through a survey of 695 undergraduate students in South Korea, we found that “privacy fatigue”—a feeling of hopelessness and weariness regarding privacy—was associated with these students’ adoption of ChatGPT. The findings revealed that this fatigue reduces perceived privacy risks, enhances their expectations of the platform, and boosts their intention to use ChatGPT. By incorporating the concepts of privacy fatigue, perceived risk, and elements from the unified theory of acceptance and use of technology, we developed a novel model to understand this phenomenon. Interestingly, despite the potential risks, the intention to use ChatGPT was not significantly influenced by perceived risk. This study contributes to a better understanding of undergraduate students’ privacy perceptions when leveraging ChatGPT in education.

Keywords ChatGPT · Privacy fatigue · Educational technology · Perceived risk · Unified theory of acceptance and use of technology · South Korea

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1 Introduction

Currently, OpenAI's generative pre-trained transformer (ChatGPT) commands widespread acclaim in various fields. It plays a significant role in cybersecurity by discerning and mitigating disinformation campaigns, identifying vulnerabilities, and formulating cybersecurity solutions (Al-Hawawreh et al., 2023). Choi et al. (2023) anticipated its potential usefulness for law school students during examinations, suggesting that its performance could potentially meet the standards for earning a Juris Doctor (JD) from a selective law school, bypassing other skill-based requirements. Li et al. (2023) demonstrated its potential for colorectal surgery, showing that it can provide tailored and precise medical information, help reduce errors and complications, and ultimately help improve patient outcomes. Thus, the versatility and broad applicability of ChatGPT have garnered positive recognition in various fields.

However, ChatGPT is also under scrutiny due to concerns about its potential threat to privacy. Its operation involves users inputting content after a prompt, to which ChatGPT responds, raising apprehensions about privacy infringements. According to the policy of OpenAI, data from user interactions are shared with third parties (Al-Hawawreh et al., 2023). Moreover, ChatGPT accumulates extensive PII from users through various channels and retains all conversational data, including message content and device details (Al-Hawawreh et al., 2023). The privacy policy of ChatGPT, over 2,000 words long, states that the sharing and accumulation of PII may occur without additional user notifications, except when mandated by legislation (Office of the Privacy Commissioner of Canada, 2023). Moreover, these privacy concerns are exacerbated by fears about “do anything now” (DAN) attacks and research pointing to backdoor attacks during parameter tuning using ChatGPT (Wang et al., 2023). Italy suspended ChatGPT for approximately four weeks, later reinstating it after implementing verification procedures to confirm the users' age and obtain consent from a parent or legal guardian if the users are under 13 years old (Schubert, 2023). Furthermore, the United States, Poland, and Canada have launched investigations into OpenAI owing to concerns regarding potential privacy breaches with ChatGPT (Charlish, 2023; Office of the Privacy Commissioner of Canada, 2023; Zakrzewski, 2023).

Privacy concerns regarding ChatGPT differ significantly from those associated with Google and Meta, which explicitly use PII for commercial purposes such as targeted advertising and profile enhancements (Fortra, 2023). For example, Google may use data from user searches to customize advertisements based on their interests, thereby refining user profiles over time. By contrast, ChatGPT mainly operates with short-term memory (Afifi-Sabet, 2024) and does not inherently require PII to function. However, temporary use of PII within sessions can occur, such as when a user inadvertently shares sensitive information during a chat. Additionally, a user may provide their email address to an integrated third-party service to receive a transcript of the conversation; however, this is optional and is not required for ChatGPT to function. Although ChatGPT functions efficiently without requiring personal data, these instances underscore the nuanced nature of its data-processing practices. Despite OpenAI's policies emphasizing anonymization and legal compliance, there remains a subtle risk of exposure that can be challenging to fully mitigate, including

the theoretical risk of data interception during transmission. The real issue with ChatGPT is that users may inadvertently share sensitive information during interactions, even without logging in or providing PII. Research (Sebastian, 2023b) has shown that users often unknowingly share personal or sensitive information when interacting with chatbots.

Although concerns regarding ChatGPT are evident, one notable demographic that seems to favor its use is undergraduate students (Cotton et al., 2023; Farrokhnia et al., 2023). With its ability to create personalized learning experiences and promote advanced learning outcomes through adaptive learning, ChatGPT has emerged as a potent tool for designing effective educational tasks (Farrokhnia et al., 2023). This facilitates experiential learning, which allows students to assess diverse strategies through game-based learning or student-centered approaches (Cotton et al., 2023). Owing to these advantages, various colleges and universities have taken measures to provide undergraduate students with usage guidelines for ChatGPT and have even established related special lectures (Baik, 2023; Ha, 2023; Lee, 2023). This attests to the fact that undergraduate students are among the groups that frequently use ChatGPT. However, despite its popularity in the educational sector, ChatGPT's exceptional performance may not be the sole reason for its widespread adoption, and concerns about privacy issues remain. Nonetheless, related information on the underlying reasons is limited—a topic that served as the starting point for our research.

We analyzed the annual privacy protection reports released by the Personal Information Protection Commission (2020, 2021, 2022); the survey data indicated that a high percentage of individuals demonstrated a passive attitude and took no action for redress following personal information leaks. This situation, characterized by a sense of passivity, leads to privacy fatigue, which refers to feelings of resignation and exhaustion regarding privacy issues (Choi et al., 2018). A certain degree of privacy fatigue may prevail among people, and undergraduate students are no exception to this. In this study, we focused on whether privacy fatigue is prevalent among undergraduate students.

To the best of our knowledge, limited research has been conducted on privacy fatigue among undergraduate students who frequently use ChatGPT. Given the increasing concerns regarding privacy and potential risks associated with ChatGPT, understanding the perceived risks related to privacy is crucial. However, a noticeable gap exists in the literature regarding the relationship between this perceived risk and privacy fatigue among undergraduate students using ChatGPT as well as the influence of these factors on its use. Ultimately, we aimed to determine whether privacy fatigue influences undergraduate students' use of ChatGPT and explore the relationship between privacy fatigue and perceived risk.

The contributions of this study can be summarized as follows. First, we introduce a novel approach that integrates privacy fatigue, perceived risk, and the unified theory of acceptance and use of technology (UTAUT). Using this approach, we analyzed the relationship between specific UTAUT elements, privacy fatigue, and perceived risk.

Second, our study identified the impact of privacy fatigue on the use of ChatGPT among undergraduate students. We propose that students experiencing privacy fatigue may be more susceptible to privacy risks, as evidenced by their increased willingness to interact with ChatGPT. This suggests a cycle where students more

accepting of privacy concerns might rely more on ChatGPT, potentially increasing privacy risks.

Third, our study revealed that the perceived risk associated with ChatGPT does not affect the intention to use it, contrary to our initial assumptions. This implies that students' focus on their performance with ChatGPT outweighs any concerns about potential risks.

2 Background

2.1 Popularity of ChatGPT in South Korea's educational sector

Currently, a ChatGPT craze is sweeping through South Korean universities. Education technology companies in the United States, such as Duolingo and Quizlet, which are popular among American high school students, have already integrated ChatGPT into their applications (Yoo, 2023). Similarly, in South Korea, there has been a movement toward leveraging ChatGPT in education. At Korea University, guidelines for utilizing ChatGPT have been established to foster education that cultivates student proficiency, encouraging its use in active learning (Ha, 2023). Additionally, special lectures explaining the operating principles of and utilization strategies for ChatGPT have been held at Chung-Ang University, providing guidance on its use. Furthermore, with the introduction of courses on research ethics and chatbots, the educational sector is witnessing a shift (Lee, 2023). Thus, in the South Korean educational context, there appears to be a prevalent voice advocating the effective utilization of ChatGPT (Baik, 2023). Therefore, to better understand the phenomenon, research on perceptions regarding ChatGPT in the educational sector would be valuable.

2.2 Digital media familiarity among South Korean students

The global trend of digitalization permeates various aspects of people's daily lives across nations, including South Korea. Known for its rapid technological advancements, South Korea has swiftly adopted digital technologies in the field of education (Lee et al., 2009). This makes South Korea an interesting setting for technological awareness surveys. Given South Korea's high student population and educational standards (Seth, 2002), investments in e-learning are regarded as highly cost-effective (Lee et al., 2009). The educational system in South Korea actively promotes online lecture learning and the use of digital media; South Korean students are thus becoming closely acquainted with digital technologies, making them ideal candidates for studies exploring perceptions regarding ChatGPT in the educational context.

3 Conceptual framework and hypotheses

3.1 Conceptual framework

The UTAUT is a comprehensive cause-and-effect model developed to explain and predict the acceptance and utilization of technology across various sectors. Many empirical studies have confirmed its validity and reliability (Foroughi et al., 2023; Khan et al., 2023; Li et al., 2024; Venkatesh et al., 2003). Given the popularity of ChatGPT among undergraduate students, our research sought to explore their use of ChatGPT using UTAUT as the theoretical framework.

The UTAUT model comprises the following factors: performance expectancy, a user's anticipation of the advantages or outcomes that they may attain by using this technology; effort expectancy, a user's perception of the level of effort required to operate the technology under investigation; social influence, a user's perceptions of the extent to which their social networks influence their technology usage; facilitating conditions, resources and support available to users for engaging with technology; behavioral intention, a user's willingness or inclination to adopt or utilize a specific technology; and use behavior, tangible actions of a user when interacting with technology.

In the exploration of the intended use of ChatGPT, attempts have been made to integrate traditional models with new elements (Foroughi et al., 2023; Strzelecki, 2023). However, to date, there has been no comprehensive research taking into account concerns about privacy and analyzing whether the intended use of ChatGPT is altered due to the abandonment or expectations thereof. To understand the impact of privacy fatigue and perceived risk among undergraduate students on their intention to use ChatGPT, we modified the UTAUT to devise a fatigue-risk-UTAUT model (Fig. 1). Based on research by Venkatesh et al. (2003), who suggested that facilitating conditions may not be significant when performance expectation and effort expect-

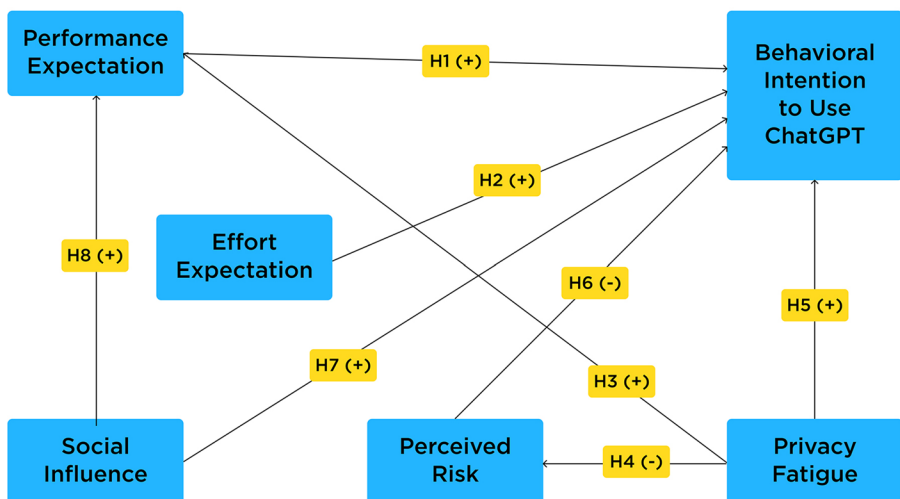


Fig. 1 Fatigue-risk-UTAUT model

tancy are present as indicators, we removed facilitating conditions to focus more on other indicators. In fact, Venkatesh et al. (2003) also noted that facilitating conditions do not directly predict behavioral intention but instead predict actual use behavior, further supporting our decision to exclude this variable.

According to the UTAUT (Venkatesh et al., 2003), performance expectancy is the strongest predictor of intention in each model and it remains significant across all measurement points. Additionally, intrinsic motivation may work through effort expectations. In other words, both performance and effort expectations positively affect behavioral intentions. This relationship has been observed in studies that investigated the acceptance attitude of ChatGPT users using the UTAUT (Duong et al., 2024; Strzelecki, 2023, as well as in research in other fields, such as LMS (Bayaga & Du Plessis, 2024). Accordingly, the following hypotheses are proposed:

H1. The performance expectations of undergraduate students positively influence their behavioral intention to use ChatGPT.

H2. The effort expectations of undergraduate students positively influence their behavioral intention to use ChatGPT.

3.2 Privacy fatigue

Privacy fatigue is defined as the growing difficulty in managing digital personal data and frequent consumer data breaches, which has left individuals feeling hopeless and fatigued about online privacy. It refers to a negative emotional state in which individuals become weary of making decisions related to their personal privacy and are reluctant to proactively oversee and regulate their personal privacy (Shao et al., 2022). Considering that the concept of “privacy” originated from Judge Thomas Cooley’s notion of “the right to be let alone” (Prosser, 1960), the phenomenon of privacy fatigue feels paradoxical in a personalized modern society. This phenomenon can be explained as follows: As information technology has advanced, data breaches have increased, initially sparking alarms, but soon leading to desensitization (Tang et al., 2021).

Conceptually, privacy fatigue involves emotional exhaustion and cynicism, leading to a shift in how individuals approach information safeguarding. Privacy cynicism, a component of privacy fatigue, reflects a skeptical and detached attitude toward privacy protection, often perceiving it as futile and unattainable (Luzsa & Mayr, 2022).

In this context, Lee et al.’s (2024) study revealed that emotional reactions to technology significantly influence user attitudes, surpassing the impact of cognitive factors such as thoughts and beliefs. The findings suggest that individuals use their emotional responses to shape their performance beliefs. Similarly, privacy fatigue, by altering emotional engagement with privacy concerns, may lead to a reallocation of cognitive resources. Individuals experiencing privacy fatigue may redirect their focus from privacy concerns to other areas where they feel they would have more control or potential for positive outcomes, thereby enhancing their performance expectations in those areas. Therefore, it can be inferred that privacy fatigue affects

emotional engagement and has a significant impact on performance expectations by shifting focus and resources to areas perceived as more manageable.

Considering the privacy calculus (Dinev & Hart, 2006), which balances the potential risks of sharing personal information against the perceived benefits, privacy fatigue might shift this balance.

Individuals suffering from privacy fatigue might perceive the emotional costs of maintaining strict privacy protection as too high and thus prioritize other aspects of their lives or work. This shift in priorities and the reallocation of cognitive resources can lead to enhanced performance in areas where individuals feel they have more control and potential for positive outcomes, as the perceived futility of privacy protection lowers the associated emotional burden and allows for greater focus on other domains.

Thus, it is reasonable to hypothesize that privacy fatigue could have a specific impact on certain populations, such as undergraduate students, who often juggle multiple responsibilities and stressors. Applying Lee et al.'s (2024) findings to the context of privacy fatigue, we can infer that emotional states play a significant role in shaping performance expectations. Therefore, we propose the following hypothesis:

H3. Privacy fatigue in undergraduate students positively influences their performance expectations.

Prior studies (Hinds et al., 2020; Shao et al., 2022; Yang et al., 2023) have suggested that privacy fatigue can lead to a sense of disengagement and decreased motivation to protect one's privacy. Privacy fatigue emerges when individuals feel inundated by continuous data breaches, resulting in diminished enthusiasm for protecting personal information and a sense of vulnerability to potential threats (Hinds et al., 2020). Moreover, when users experience fatigue, instead of actively pursuing methods to safeguard their privacy, they tend to withdraw, show signs of skepticism, and feel emotionally drained because of concerns about data breaches and privacy threats (Shao et al., 2022; Yang et al., 2023).

This disengagement and decreased motivation can cause individuals to perceive privacy risks as less significant, as they become desensitized to potential threats.

Given that undergraduate students often face considerable academic and social pressure, the impact of privacy fatigue on this group may be particularly pronounced. As they juggle various responsibilities, the additional burden of managing privacy concerns may lead to a greater sense of fatigue, and consequently, a reduced perception of privacy risks. This is because their cognitive and emotional resources are already stretched thin, making it more difficult for them to actively engage with and assess privacy threats.

H4. Privacy fatigue in undergraduate students negatively influences their perceived risks.

According to Choi et al. (2018), people who experience significant privacy fatigue are less inclined to deliberate on how and if they should share personal details, and they may consent to sharing their information without much thought. Despite the

ongoing concerns raised about ChatGPT, individuals who consent to sharing their information without much consideration are likely to be indifferent to ChatGPT's privacy issues and have a higher intention to use ChatGPT.

Given that undergraduate students are often early adopters of new technologies and frequently engage with digital platforms for both academic and social purposes, their experience of privacy fatigue may similarly influence their behavior. As privacy fatigue sets in, these students may become less concerned about the privacy implications of using ChatGPT and more inclined to use it without much hesitation.

H5. Privacy fatigue in undergraduate students positively influences their behavioral intention to use ChatGPT.

3.3 Perceived risk

Perceived risk refers to individuals' subjective evaluation of potential negative outcomes, encompassing performance, financial, time, psychological, and privacy risks (Han et al., 2024). The notion of perceived risk plays a pivotal role in diverse domains such as marketing, finance, and cybersecurity, because individuals often base their choices on it (Emami-Naeini et al., 2021). In cybersecurity, the concept of perceived risk is used to understand how individuals perceive vulnerabilities (Debb & McClellan, 2021), how marketers view cybersecurity risks (Mathur, 2018), the influence of heuristics on risk perception (Van Schaik et al., 2020), and the relationship between risk perception and precautionary behavior (Van Schaik et al., 2017). This metric is commonly used; importantly, it can serve as a significant measure for expressing the degree of risk perception of undergraduate students regarding ChatGPT.

A prior study (Murtarelli et al., 2023) investigated the quality of chatbots in the fashion industry and discovered that perceived risk has a negative impact on the behavioral intention to use them. Another study (Mehroliya et al., 2023) aimed to understand customers' perceived risk related to banking chatbots, which was found to have a negative impact on their continued use. These findings suggest that perceived risk can negatively influence the behavioral intention to use chatbots across different domains.

Given that undergraduate students frequently use digital platforms for both academic and social purposes, their perceived risk regarding ChatGPT is particularly relevant. If these students perceive significant risks associated with using ChatGPT, such as privacy concerns or data breaches, this perception may negatively impact their intention to use the platform.

H6. Undergraduate students' perceived risk negatively influences their behavioral intention to use ChatGPT.

3.4 Social influence

Social influence is conceptualized as the existence of various social norms centered on normative beliefs that shape a given behavior (Joa & Magsamen-Conrad, 2021). This influence is crucial in shaping an individual's technology adoption and usage behavior. Peer pressure, a specific type of social influence, plays a significant role in this process. A study (Sahu, 2022) conducted with undergraduate students at Tezpur University in India, a country with a large youth population, confirmed that peer pressure influences undergraduate students' behaviors, values, and attitudes. Additionally, in Nigeria, the impact of peer pressure was found to be high among undergraduate students, significantly influencing tobacco use. Therefore, undergraduate students are influenced by peer pressure, and considering the nature of social influence, it can be inferred that they are subject to the broader effects of social influence.

Given that peer pressure is a subset of social influence, it would be logical to extend these findings to other forms of social influence. The influence of others has been found to play a pivotal role in deciding whether to use ChatGPT for educational purposes (Yilmaz et al., 2023), as evidenced by a recent study conducted at a university in Nepal using the UTAUT2 model, which demonstrated that social influence had a significant effect on students' behavioral intention to use ChatGPT (Bahadur et al., 2024). Bayaga and Du Plessis (2024) investigated the behavioral intentions of university staff toward the use of a learning management system (LMS) and found that social influence significantly affected the behavioral intention to adopt it. Furthermore, it has been proven in various contexts (e.g., mobile learning and e-learning platforms) that social influence is crucial for determining users' behavioral intentions regarding technology adoption (Strzelecki, 2023). Therefore, it can be inferred that social influence among undergraduate students positively influences their intention to use ChatGPT.

H7. The social influence of undergraduate students positively influences their behavioral intention to use ChatGPT.

Social influence may also affect performance expectations. It plays a pivotal role in elevating users' initial confidence in chatbots, which in turn amplifies their inclination to utilize chatbots and fosters active interactions (Choudhury & Shamszare, 2023). The concept of confirmation bias, derived from the renowned four-card task (Evans, 2016) and the finding that people tend to seek information that confirms rather than refutes their pre-existing assumptions, implies that an individual's judgment influences their expectations of the outcome. Based on these theories, it was inferred that social influence affects expectations.

H8. The social influence of undergraduate students positively influences their performance expectations.

4 Methodology

4.1 Instrument development

The instrument was developed by measuring all items on a 5-point Likert scale ranging from “strongly disagree” to “strongly agree.” All items were considered reliable and sourced from various studies (Pavlou et al., 2007; Schaufeli et al., 1996; Venkatesh et al., 2003). They are widely regarded as exemplary reference models for measuring related attributes. We integrated the key components of our study, namely “undergraduate students” and “ChatGPT,” into these items. The items concerning performance expectation (PE), effort expectancy (EE), social influence (SI), and behavioral intention to use ChatGPT (BI) were adopted from a study by Venkatesh et al. (2003). The items related to privacy fatigue (PF) were adapted from Schaufeli et al.’s (1996) study, and those associated with perceived risk (PR) were drawn from Pavlou et al.’s (2007) study. The original items referenced from other studies were in English and were translated by a bilingual English-Korean/Korean-English translation expert and underwent notarization to enhance the reliability of the survey process. Table 1 lists the items used in the study.

Given the potential for skewed survey data, we employed partial least squares structural equation modeling (PLS-SEM) for both the pilot and main tests. This method was selected because of its nonparametric.

Table 1 Items and sources

Variable	Item	Subcode	Source
PE	I find that using ChatGPT helps my campus life.	PE1	Venkatesh et al. (2003)
	Using ChatGPT enables me to accomplish tasks more quickly.	PE2	
	Using ChatGPT increases my productivity.	PE3	
EE	It would be easy for me to become skillful at using ChatGPT.	EE1	
	I find ChatGPT easy to use.	EE2	
	Learning to operate ChatGPT is easy for me.	EE3	
SI	People who influence my behavior think that I should use ChatGPT.	SI1	
	People who are important to me think that I should use ChatGPT.	SI2	
	In general, students at my university have positive views on using ChatGPT.	SI3	
BI	I predict I would use ChatGPT in the next 6 months.	BI1	
	I intend to use ChatGPT in the next 6 months.	BI2	
	I plan to use ChatGPT in the next 6 months.	BI3	
PF	It is tiresome for me to care about privacy while using ChatGPT (Removed after the pilot test due to a low outer loading of 0.415).	PF1	Schaufeli et al. (1996)
	I have become less interested in privacy issues while using ChatGPT.	PF2	
	I just want to use ChatGPT and not to be bothered by privacy issues.	PF3	
PR	I am afraid that someone else may review my queries that include personal information.	PR1	Pavlou et al. (2007)
	I think that ChatGPT has a risk to accumulate my data (Removed after the pilot test due to a low outer loading of 0.622).	PR2	
	I am concerned that ChatGPT may disclose my personal data.	PR3	

nature, which provides flexibility in model estimation with various data distributions (Hair et al., 2019). To improve the generalizability of our findings, the model was evaluated using bootstrapping techniques and employing 5000 resampling iterations.

A preliminary questionnaire was presented and discussed with associates. A revised version of the questionnaire was developed based on the feedback received, specifically regarding the items that best represented the variables. The revised version was subsequently pilot-tested in November 2023 on an online community platform at Korea University involving 107 undergraduate students. This sample size aligns with the recommended range of 100–200 participants for instrument development (Kline, 2014). Through a pilot test, we verified the reliability and validity of the structural model based on these items. The second item related to perceived risk (PR2) had an outer loading value of 0.622, and the first item related to privacy fatigue (PF1) had an outer loading value of 0.415. These values were below the recommended threshold. According to Hair et al. (2021), the outer loading value should exceed 0.7 to ensure model reliability. Therefore, we removed these items. Moreover, it is recommended that the heterotrait–monotrait ratio (HTMT) value be lower than 0.85 (Franke & Sarstedt, 2019), and all our items showed values that did not exceed this threshold. Additionally, the square root of the average variance extracted (AVE) was greater than the correlation coefficient for all items. Thus, discriminant validity was verified to be high (Fornell & Larcker, 1981), ensuring that one item was not confused with another. Subsequently, we conducted the main survey.

4.2 Sample

Data collection for the main survey, targeting undergraduate students in South Korea, was conducted in November 2023. It was conducted online by Macromill Embrain, a research firm, using its panel system. Initially, participants were asked several screening questions (Q1–Q8 in the Online Resource) to determine their eligibility for the study. Only undergraduate students who had prior experience using ChatGPT and voluntarily agreed to participate in the research were considered qualified at this initial stage, and they were subsequently presented with the main survey questions. After obtaining ethical approval, a survey was administered to 700 undergraduate students. We chose this sample size in consideration of recent studies investigating the perceptions related to ChatGPT adoption, most of which had sample sizes ranging from high-300s to mid-500s (Foroughi et al., 2023; Sallam et al., 2023; Strzelecki, 2023; Tiwari et al., 2024). We aimed to enhance the reliability of our survey; thus, we opted for a larger sample size than that considered in previous studies.

We specifically excluded those who had only used services incorporating ChatGPT features, as our primary focus was on individuals with direct experience using ChatGPT. Of the 700 participants, five were excluded, resulting in a final sample of 695 individuals for analysis. The final sample consisted of 251 male and 444 female respondents. The age group closest to the average was 22–23 years and the duration of usage closest to the average was 3–4 months. Table 2 provides an overview of the sample demographics with percentages rounded to the second decimal place. We

Table 2 Demographic profile ($N=695$)

Measure	Item	Count	%
Age (years)	18–19	84	12.09
	20–21	189	27.19
	22–23	231	33.24
	24–25	122	17.55
	≥ 26	69	9.93
Duration of use (months)	< 1	163	23.45
	1–2	120	17.27
	3–4	130	18.71
	5–6	133	19.14
	7–8	58	8.35
Gender	≥ 9	91	13.09
	Male	251	36.12
	Female	444	63.88

Table 3 Reliability and validity metrics

Variable	Subcode	Construct reliability and validity				Discriminant validity (HTMT)					
		α	CR (rho c)	AVE	Outer loading	BI	EE	PE	PF	PR	SI
BI	BI1	0.946	0.965	0.902	0.943	0.334					
	BI2				0.96						
	BI3				0.946						
EE	EE1	0.803	0.885	0.719	0.783	0.334					
	EE2				0.88						
	EE3				0.877						
PE	PE1	0.803	0.884	0.717	0.858	0.679	0.358				
	PE2				0.852						
	PE3				0.83						
PF	PF2	0.617	0.839	0.723	0.863	0.379	0.3	0.378			
	PF3				0.838						
PR	PR1	0.866	0.937	0.882	0.937	0.077	0.107	0.075	0.486		
	PR3				0.941						
SI	SI1	0.773	0.87	0.691	0.865	0.544	0.306	0.701	0.387	0.057	
	SI2				0.878						
	SI3				0.744						

utilized these datasets for the first stage of PLS-SEM, specifically for measurement model assessment, for reliability, validity, and discriminant validity testing.

4.3 Reliability and validity

The structural model satisfied all criteria for reliability and validity. To ensure the consistency and explanatory power of the measurement items for each variable, we assessed the Cronbach's alpha (α), composite reliability (CR), AVE, and outer loadings, as presented in Table 3. All relevant variables exhibited an α -value greater than 0.6, which indicates an acceptable reliability as noted in Shi et al.'s (2012) study. Furthermore, the measurement model exceeded 0.8, surpassing the recommended 0.7 threshold for CR (Taber, 2018). Similarly, the model consistently met the benchmark

set by prior research with a minimum AVE value of 0.691, exceeding the suggested 0.5 threshold (Fornell & Larcker, 1981). Additionally, we confirmed that the outer loading value of the main model exceeded 0.7 (Hair et al., 2021), satisfying this crucial criterion.

4.4 Discriminant validity

To assess whether each variable was distinct from the others, thereby establishing discriminant validity, we reviewed the HTMT values and compared the square root of the AVE for each variable with its correlations with other variables. Although methods such as the Fornell–Larcker criterion, cross-loadings, and variance inflation factor (VIF) can be employed to assess discriminant validity, we chose the HTMT because of its advantages in this case. Both the Fornell–Larcker criterion and cross-loadings rely on correlations, potentially missing complex or nonlinear relationships. Furthermore, unlike the VIF, the HTMT directly evaluates the actual relationships between constructs, rendering it less vulnerable to multicollinearity. Table 3 shows that the HTMT values in our model fall below 0.85, which is the threshold considered indicative of sufficient discriminant validity (Henseler et al., 2015).

As shown in Table 4, the square root of the AVE surpasses the correlation coefficient for all constructs.

According to Fornell and Larcker (1981), the discriminant validity of a construct is supported when the square root of the AVE exceeds its correlation with other constructs. In other words, the square root of the AVE for each construct should be greater than that of the non-diagonal elements in the corresponding row and column. Our model consistently satisfies this criterion.

4.5 Data analysis

We constructed a path model using SmartPLS version 4.0.9.6 (Ringle et al., 2022). As part of the second phase of the PLS-SEM approach (i.e., structural model assessment), we scrutinized the structural model by analyzing the path coefficients (β) and P -values (P) of our proposed model.

According to the PLS-SEM fit metrics, the model displays a good fit. The saturated model, which represented the full model with all possible relationships between the variables, yielded a standardized root mean square residual (SRMR) of 0.06. In contrast, the estimated model, which was derived from the sample data and represented the proposed relationships between the variables, had an SRMR of 0.07. Both

Table 4 Correlation matrix of constructs excluding diagonal values with the square root of AVE (in bold)

	BI	EE	PE	PF	PR	SI
BI	0.949					
EE	0.291	0.848				
PE	0.594	0.286	0.847			
PF	0.288	0.211	0.265	0.85		
PR	-0.07	-0.09	-0.055	-0.359	0.939	
SI	0.469	0.242	0.557	0.266	-0.048	0.831

values fall below the reference benchmark of 0.08 (Hu & Bentler, 1999). The normed fit index (NFI) for the saturated model is 0.79 and 0.80 for the estimated model. It is commonly understood that an NFI value closer to one indicates a more optimal fit (Bentler & Bonett, 1980). Given that all index values align with the reference standards, we can confidently infer the relationships between the variables. Figure 2 presents a graph that illustrates the PLS-SEM model.

4.6 Ethical considerations

This study was approved by the Institutional Review Board (IRB) at Korea University and the approval number is KUIRB-2023-0386-01.

We made every effort to treat survey participants with due regard and respect. The participants were presented with an online research consent form, which explained that participation in the study was voluntary and that if they chose not to participate, they could decline at any time. Additionally, the form provided contact information for any questions regarding the ethical considerations or rights of the participants. The survey commenced only after the participants provided their informed consent.

5 Results and discussion

5.1 Hypothesis testing

Setting aside assumptions regarding moderation effects, all proposed hypotheses were supported, except for H6 (Table 5).

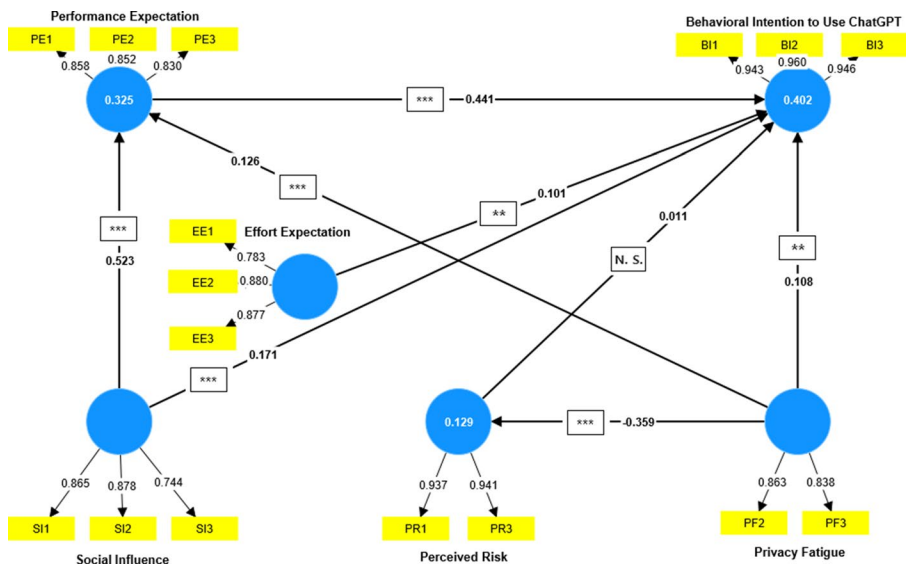


Fig. 2 PLS-SEM model. Note: *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, N. S. (not significant): $P > 0.05$

Table 5 Path coefficients and hypothesis testing

Path	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T-statistic ((O/STDEV))	P-value	Hypothesis	Result
PE → BI	0.441	0.442	0.042	10.513	0	H1 (+)	Supported
EE → BI	0.101	0.102	0.034	3	0.003	H2 (+)	Supported
PF → PE	0.126	0.127	0.031	3.998	0	H3 (+)	Supported
PF → PR	-0.359	-0.359	0.036	9.833	0	H4 (-)	Supported
PF → BI	0.108	0.108	0.034	3.187	0.001	H5 (+)	Supported
PR → BI	0.011	0.01	0.03	0.352	0.725	H6 (-)	Not supported
SI → BI	0.171	0.17	0.04	4.29	0	H7 (+)	Supported
SI → PE	0.523	0.524	0.03	17.688	0	H8 (+)	Supported

Performance expectancy strongly influences the behavioral intention to use ChatGPT ($\beta=0.441$, $P<0.001$), supporting H1. We also found that effort expectancy positively influences the behavioral intention to use ChatGPT ($\beta=0.101$, $P<0.01$), supporting H2. Additionally, the analysis revealed that privacy fatigue has a positive impact on performance expectations ($\beta=0.126$, $P<0.001$), confirming H3. Moreover, privacy fatigue negatively affects the perceived risk of using ChatGPT ($\beta=-0.359$, $P<0.001$), supporting H4. We also found that privacy fatigue has a positive impact on the behavioral intention to use ChatGPT ($\beta=0.108$, $P<0.01$), supporting H5. Contrary to expectations, perceived risk does not significantly influence behavioral intention ($P=0.725$), rejecting H6. We also discovered that social influence positively affects the behavioral intention to use ChatGPT ($\beta=0.171$, $P<0.001$), confirming H7. Finally, social influence has a substantial positive effect on performance expectancy ($\beta=0.523$, $P<0.001$), supporting H8.

As presented in Fig. 2, our research model explains 40.2% of the variance in the behavioral intention to use ChatGPT and 32.5% in the performance expectancy of ChatGPT, as reflected by the R-squared values. Both of these values are high compared to those of previous studies (Alrawi et al., 2020; Foughi et al., 2023; Jaradat & Al Rababaa, 2013; Kaba & Touré, 2014). This suggests that our proposed framework adequately captures the variance in both the behavioral intention to use ChatGPT and performance expectancy.

5.2 Theoretical implications

Since the initial release of ChatGPT on November 30, 2022 (Balu & Sriram, 2023), there has been a continuous stream of research on user technology acceptance and perceptions of ChatGPT (Choudhury & Shamszare, 2023; Duong et al., 2024; Firat, 2023; Foughi et al., 2023; Menon & Shilpa, 2023; Sallam et al., 2023; Strzelecki, 2023; Tiwari et al., 2024). However, to our knowledge, research addressing privacy

fatigue and the perceived risk related to ChatGPT has been limited. A significant research gap persists in understanding why, despite the ongoing privacy concerns raised about ChatGPT (Al-Hawawreh et al., 2023; Chowdhury et al., 2023; Hunter, 2023; Li et al., 2023; Oviedo-Trespalacios et al., 2023; Paul et al., 2023; Sebastian, 2023a; Wach et al., 2023), undergraduate students continue to use it and how privacy fatigue and perceived risks influence their usage. This study applied the UTAUT model to assess privacy fatigue and perceived risk among undergraduate students using ChatGPT and presents important theoretical and practical implications based on the results.

First, it identified the role of privacy fatigue in undergraduate students' use of ChatGPT, revealing that students overwhelmed by privacy fatigue may be more susceptible to privacy vulnerabilities. Notably, as corroborated by H5, those experiencing privacy fatigue demonstrate a heightened intention to use ChatGPT. This trend suggests the existence of a feedback loop—a cycle in which one event reinforces and amplifies subsequent events (Ford, 1999). Students who are resigned due to privacy concerns may become increasingly reliant on ChatGPT, leading to potential privacy risks. This exposure could diminish their concerns and deepen their dependence on these platforms. In alignment with H3, these students also exhibited elevated performance expectations from ChatGPT, possibly viewing it as a solace for their privacy-related apathy. This mirrors the findings of previous studies (Kim et al., 2019; Menon & Shilpa, 2023; Schomakers et al., 2021; Xu et al., 2014) suggesting that individuals are willing to trade some level of privacy to benefit from innovative technologies. Moreover, according to H4, these students perceived fewer risks. Their growing indifference toward privacy safeguards diminishes their risk awareness, making them more vulnerable to privacy threats. The incident where the data analytics firm Cambridge Analytica misused Facebook users' personal information is a prime example of how low-risk awareness can lead to privacy threats. Previous studies (Berghel, 2018; Hinds et al., 2020; Wylie, 2019) have indicated that users grant application access permissions without sufficiently understanding how their data would be used and shared, leading to inappropriate data collection.

Second, contrary to the initial expectation that perceived risks would reduce the intention to use ChatGPT, we observed that these risks did not significantly affect this intention. This may be attributed to the high performance expectations associated with ChatGPT, along with its substantial social influence. Previous studies (Bayaga & Du Plessis, 2024; Chang et al., 2016; Khan et al., 2023) have noted similar patterns, in which low effect sizes for a specific path can be explained by the over-riding effect of other factors. The elevated performance expectations surrounding ChatGPT, coupled with the positive feedback undergraduate students received from their peers, suggest that these factors counterbalance the influence of perceived risks on the intention to use ChatGPT.

Third, we identified positive relationships between social influence, performance expectations, effort expectations, and behavioral intention to use ChatGPT. This not only sheds light on previously unexplored connections but also reaffirms the findings of existing research. For undergraduate students, the expectation of performance was the most significant contributor to their intention to use ChatGPT. This relationship has been observed in previous studies involving undergraduate students (Duong

et al., 2024; Menon & Shilpa, 2023; Shahsavar & Choudhury, 2023). Furthermore, this expectation is largely shaped by the prevailing social dynamics among these students, which have not been fully addressed in previous studies. The finding that social influence has a significant impact on the use of ChatGPT among undergraduate students supports our initial hypothesis that peer pressure within this cohort significantly influences ChatGPT use, highlighting its importance as a motivational factor (Abdaljaleel et al., 2023; Choudhury & Shamszare, 2023; Menon & Shilpa, 2023). Given the significant impact of social and performance expectations on the use of ChatGPT by undergraduate students, it is crucial to consider these factors when addressing privacy concerns. This study also found that user perceptions of ChatGPT's ease of use positively affect its usage, which aligns with prior results (Duong et al., 2024; Menon & Shilpa, 2023).

5.3 Practical implications

First, considering that privacy fatigue combined with high performance expectations for ChatGPT may amplify its usage, practical measures are needed to address students' increased vulnerability and diminished awareness of privacy risks due to privacy fatigue. OpenAI can consider assessing boundaries for data collection concerning undergraduate students' privacy, and ideally offer them a choice to determine the extent of data collected. The need for such measures was underscored by the Cambridge Analytica incident involving a Facebook data breach. Facebook provides users with the option to decide which applications and sites can be connected to their Facebook accounts (Browne, 2018). However, it is essential that notifications regarding data collection are presented in a clear, easily understandable, and user-centric manner. Moreover, to enhance undergraduate students' understanding of privacy risks, an effective strategy may be to embed ChatGPT's privacy-related content in regular undergraduate courses. According to May and Luth (2013), integrating specific educational content into standard courses, rather than standalone lectures, can lead to more positive behavioral outcomes.

Second, given the significant influence of social pressure on the use of ChatGPT and its performance expectations, implementing a social media campaign may effectively raise awareness regarding privacy risks. Various studies (Al-Dmour et al., 2020; Momany & Alshboul, 2016; Oyza & Edwin, 2015) have demonstrated the effectiveness of social media campaigns in leveraging social influence. However, it is crucial that such campaigns do not overwhelm students with excessive information, which may easily be disregarded. Instead, presenting concise and attention-grabbing information may reduce the likelihood of privacy fatigue.

5.4 Limitations and future research

Our study aimed to investigate the factors influencing the use of ChatGPT, with a specific focus on undergraduate students who are enthusiastic about this technology. Although our study targeted this demographic, other groups such as computer programmers and graduate students could also be potentially avid users of ChatGPT. Future research may benefit from a broader scope encompassing these diverse groups.

Additionally, the survey was limited to South Korean undergraduate students. Furthermore, the survey agency's system was rigidly set to the binary response options of male and female under participant demographics, preventing us from asking if an individual identified as non-binary; this aspect must be addressed in future research. Future studies could also extend their reach to culturally distinct countries or include undergraduate students from multiple nations. Finally, given that the ChatGPT beta version was released less than a year before our survey, most of the respondents did not use it for an extended period. Thus, future studies should focus on users with more experience with ChatGPT, particularly as the platform matures.

Acknowledgements The authors would like to express their heartfelt gratitude to everyone who made the completion and publication of this paper possible. Special thanks go to the Editor-in-Chief, the editorial team, anonymous reviewers, and language proofreaders, whose commitment and efforts were instrumental in bringing this work to fruition.

Funding The authors did not receive support from any organization for the submitted work.

Data availability Data will be made available on reasonable request.

Declarations

Ethics approval This study was approved by the IRB at Korea University (approval number KUIRB-2023-0386-01).

Consent Survey participants gave their informed consent before participation.

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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